



Design and Development of Real-time landslide early warning system through low cost soil and rainfall sensors

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ABSTRACT

Development of cost effective early warning systems is of paramount importance for real time monitoring of natural hazards such as Landslides, Floods, Earthquake etc. The present research work discusses about the design and development of cost effective early warning system for monitoring shallow debris landslides. The warning system consist of field unit and remote network interface. The field system includes the rainfall, temperature, humidity, soil temperature, soil moisture and Tiltmeter as primary sensors for monitoring the landslide movements with an independent power supply of 50 W solar panel. The datasets that are received are stored in the local data logger with the help of microsd card. Real time monitoring of the datasets and analysis was made possible by storing the datasets in remote Database Management System. These datasets are transmitted via the Global System for Mobile communication module installed at the field unit. In order to convey the alerts during emergency time a simple visual message has been adopted for the current model. Level I, Level II and Level III, Level I being the threshold values are within the desired limit and actions are not necessary and Level III denotes threshold values have exceeded the set limit and warning are issued to local stake holders. A pilot study of the Early Warning System has been installed on site for quality checks of the sensor & data and the results were monitored. The model developed not only can address the early warning signs of mass movement but also can be calibrated to predict flood in major streams based on rainfall intensity.

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1. Introduction

Himalayan region is one of the youngest mountain region in the world with diverse landscapes and Eco-Tourism activity visited by large number of tourist every year. The region is morphologically unstable and prone to various hazards due to seasonal rainfall precipitation, torrential rainfall and tectonic activities etc. Among the natural hazards landslide are more frequent in nature and causes more social as well as economical losses in the Himalayan region [1]. The landslides are more common along the transportation corridors in the hilly regions (Fig. 1) due to inadequate geological and geotechnical investigation and improper designs of the support structures [2]. The 2017 Dhalli and Kotropi landslide that occurred in the Shimla and Mandi districts with huge social and economic losses are a prime example of landslide activities in the Himalayan region. These landslides are a constant threat to the socioeconomic

development as well as the communities residing in the hilly region. In a report published by GSI 13% of the land surface in India are prone to landslides with a loss of 4.5 billion\$ [3]. Various causative factors of landslide initiation, susceptibility and stability have been studied by researchers all over the world and especially in the Himalayan region [4–8]. In regions such as these where landslides are more frequent in the transportation corridors and settlements it is not only enough to provide to assessment and mitigation of the failed slopes but also to constantly monitor to provide warnings on imminent slope failure during monsoon season and other hazards [9]. The landslide Early Warning System (EWS) is one of the solutions employed to predict the mass movement initiation or and slope failure in areas such as stabilization of landslides are difficult and places where landslide are more frequent and consistent. In Such cases development of cost effective EWS provides a huge capability in monitoring landslides [10]. EWS has several definitions coined by different researchers. [11] coined them as “monitoring devices designed to avoid, or at least to minimize the impact imposed by a threat on humans, damage to prop-

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Fig. 1. Landslide at various locations of Himalayan region (Source: Filed Photographs).

erty, the environment, or/and to more basic elements like livelihoods and for United Nations International Strategy for Disaster Reduction they are the set of capacities needed to generate and disseminate timely and meaningful warning information to enable individuals, communities and organizations threatened by a hazard to prepare and to act appropriately and in sufficient time to

reduce the possibility of harm or loss". In recent years many types EWS has come into existence that focus on various aspects of landslide studies. EWS monitoring rainfall intensity or soil moisture content or Geospatial based EWS etc. [10] used wire Extensiometer, rain gauge and cameras to monitor the landslide in central Italy.[12,13] developed GIS based landslide warning system based

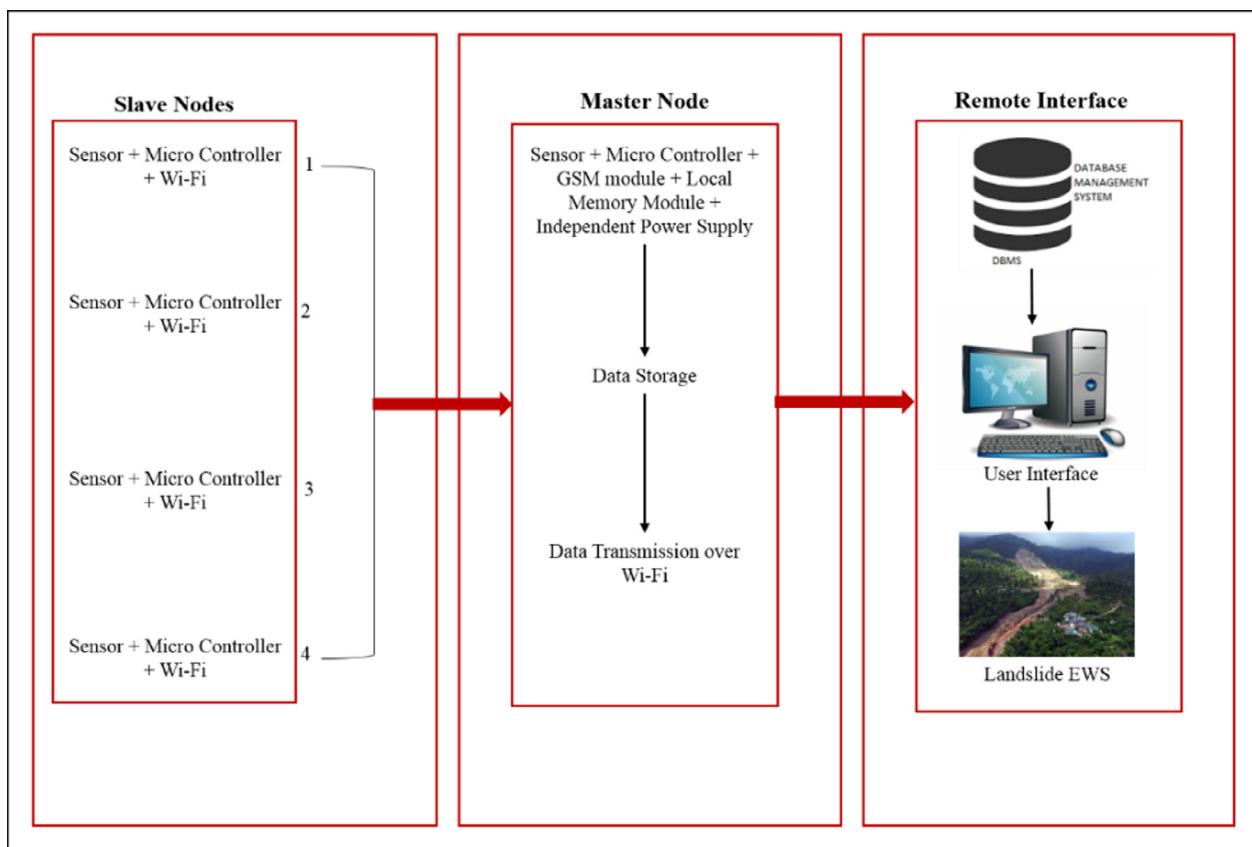


Fig. 2. Block Diagram of the Field System and Remote Interface.

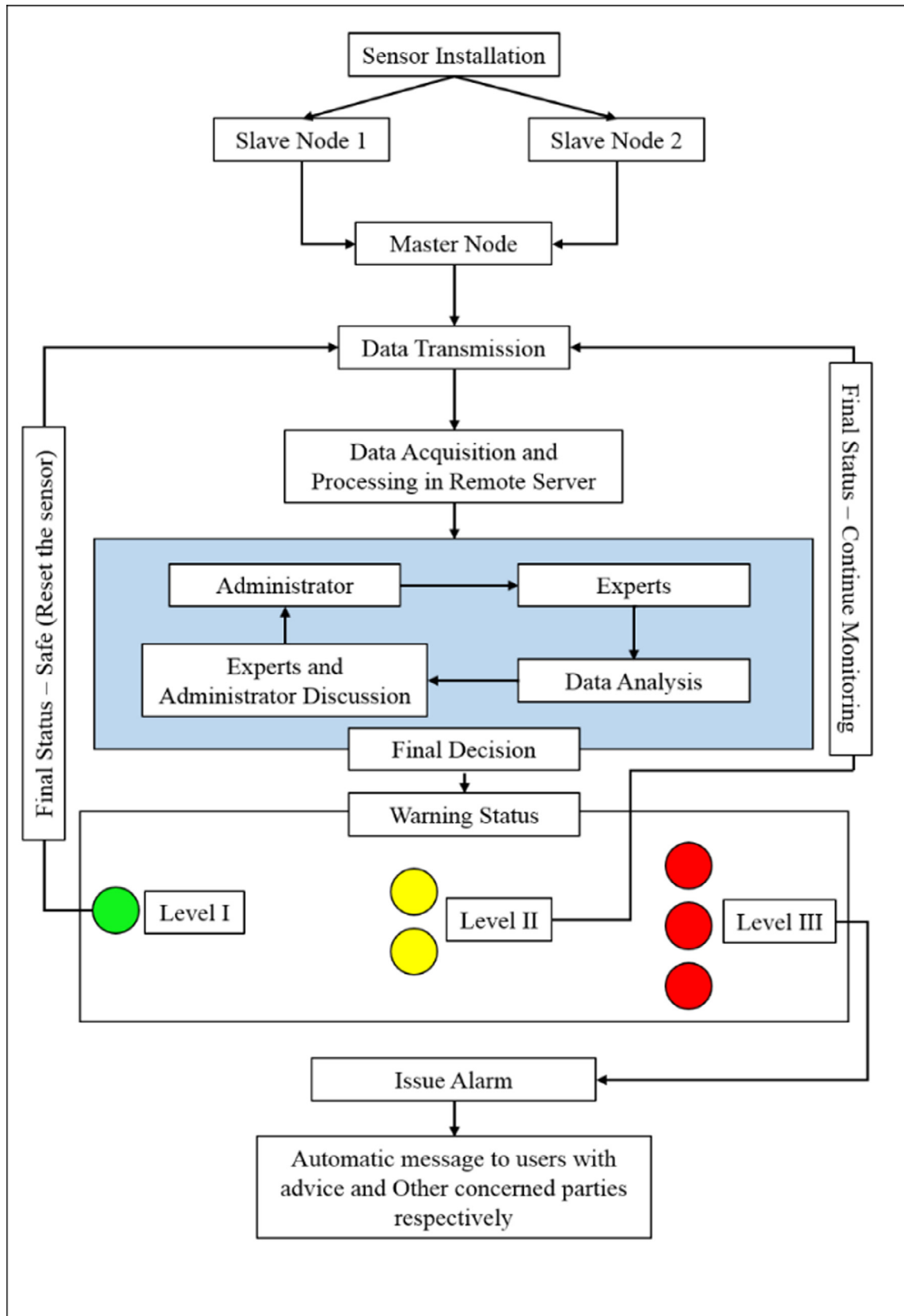


Fig. 3. Overall design approach of the Early Warning System (EWS).

on the data compiled from meteorological models, weather radars and satellite estimates. Landslide prone regions were estimated based on these datasets and used as an early warning system.

[14–16] developed early warning system based on vibration sensors, rainfall threshold and Internet of Things. Researchers from various aspects of the landslide have tried to identify and develop

hazard models and stabilization measures across the world [17–20].

Our method encompasses the use of soil, rainfall and wind datasets to develop threshold frequencies for landslide initiation specifically for landslide composed soil and rock materials. These EWS system can be deployed in remote regions with solar based power grid for continuous monitoring and data can be monitored and analysed in real-time. The current research work discusses about the design and development of cost effective early warning system for monitoring landslide movements.

2. Methodology

The first step in monitoring landslides and prediction of landslides is to identify key areas that are highly prone to land subsidence. Factors such as Soil, Geology, Lithology and Geomorphology etc. are derived from various resources to estimate landslide vulnerable areas. These data will be used in conjunction with the demographic and social data to delineate the high risk areas from other sites. The results provide details on places that requires immediate attention for installation of prototype early warning systems. Landslides in Himachal Pradesh are mainly caused by prolonged and antecedent rainfall during the month of July to December. This may be due to pore water pressure in the soils, overburden of rocks etc. The landslides occurred in and around the study area will be derived from various satellite imageries and cross correlated with their rainfall data to identify the specific threshold value at which subsidence occurs. The value may be specific to different landslides but an overall value can be summarized by studying the landslides. Other than rain gauge stations, Tiltmeter was used to detect surface movements. Sensors used to monitor the soil temperature, moisture and conductivity will be deployed along the subsurface part of the landslides to monitor the ground deformation. Data's from these instruments can be combined with rainfall data to produce accurate results of ground deformation. Real time monitoring of landslide involves various sections of analysis. The data are recorded by the sensor, stored in the local memory module and transmitted through wireless data transmission module to the remote location for real time analysis. The data from various sensor are received to a remote computer. These data are then stored in a DBMS for further analysis. The model created process the data at 30 mins interval 24*7 (Fig. 2). The results retrieved through the model can be used to map the active real time deformation in the study area. Once the values reach the designated threshold the system will be able to disperse information regarding landslide warning through various means of communication such as SMS alerts, emails, visual and auditory warning signs etc to various governmental organizations, NGO'S and local people during emergency times (Fig. 3).

3. Results and discussion

The EWS consist of two units' field based and remote interface (Fig. 4). Field Unit consist of Independent Power Supply using 50 W solar panel, Rainfall Sensor, Temperature, Humidity, Wind speed, wind direction, Soil moisture, temperature and Tiltmeter sensor. Soil sensors were installed at 45 cm below the surface to measure the saturation rate and conductivity. Bucket method rain gauge provides cheap and high accuracy data for rainfall forecast. The rainfall sensors were calibrated to provide data at 30 min interval. Other sensors namely Temperature, Humidity, Wind speed and direction were used as secondary datasets for analysis purpose. The data retrieved from the sensors were stored in a local data logger attached to master control node. The data logger uses GSM module and slot for microsd card for data storage and transmission

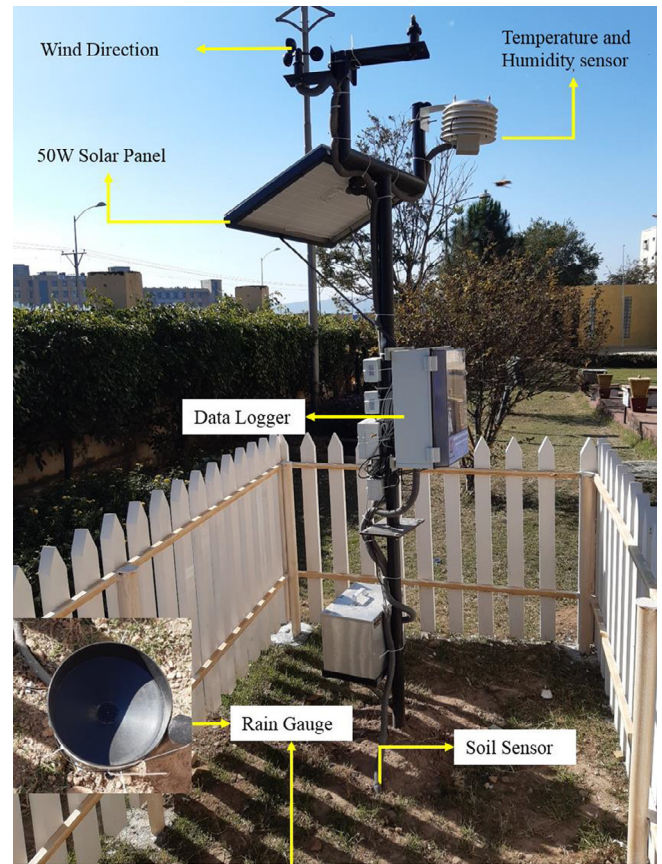


Fig. 4. Field installation of EWS.

purposes. These mechanism allows to record and store the data during a crisis such as network problem and transmit the data when the connection has been retailed. The GSM module has been configured in a way to work with existing sim cards via 3G/4G/5G networks. The datasets are then transmitted and stored on a DBMS in a remote server. These datasets will be monitored real time for landslide movements to provide early warning signs of slope failure.

In our methodology we adopted automated as well as knowledge driven method to evaluate the results of the data. During emergency times the datasets will be evaluated in-house through researchers and scientist before issuing the alert. In order to make the method spare we adopted simplicity as criteria. In emergency the information's must be simple and straight forward for all the stake holder to understand that includes the local authorities,

Table 1
Classification levels of alert messages for the Early Warning system.

Warning Level	Trigger	Reason	Response
Level I	Default	All threshold values are within safe limit	Continuous daily monitoring
Level II	Rainfall	Rainfall threshold values exceed	Datasets were continuously analysed by experts to issue the warning level to various stake holders
Level III	Rainfall, Tiltmeter and Soil Sensors	Threshold values have exceeded the desired limit	Datasets were continuously analysed by experts and warning are issued to local authorities through bulletins

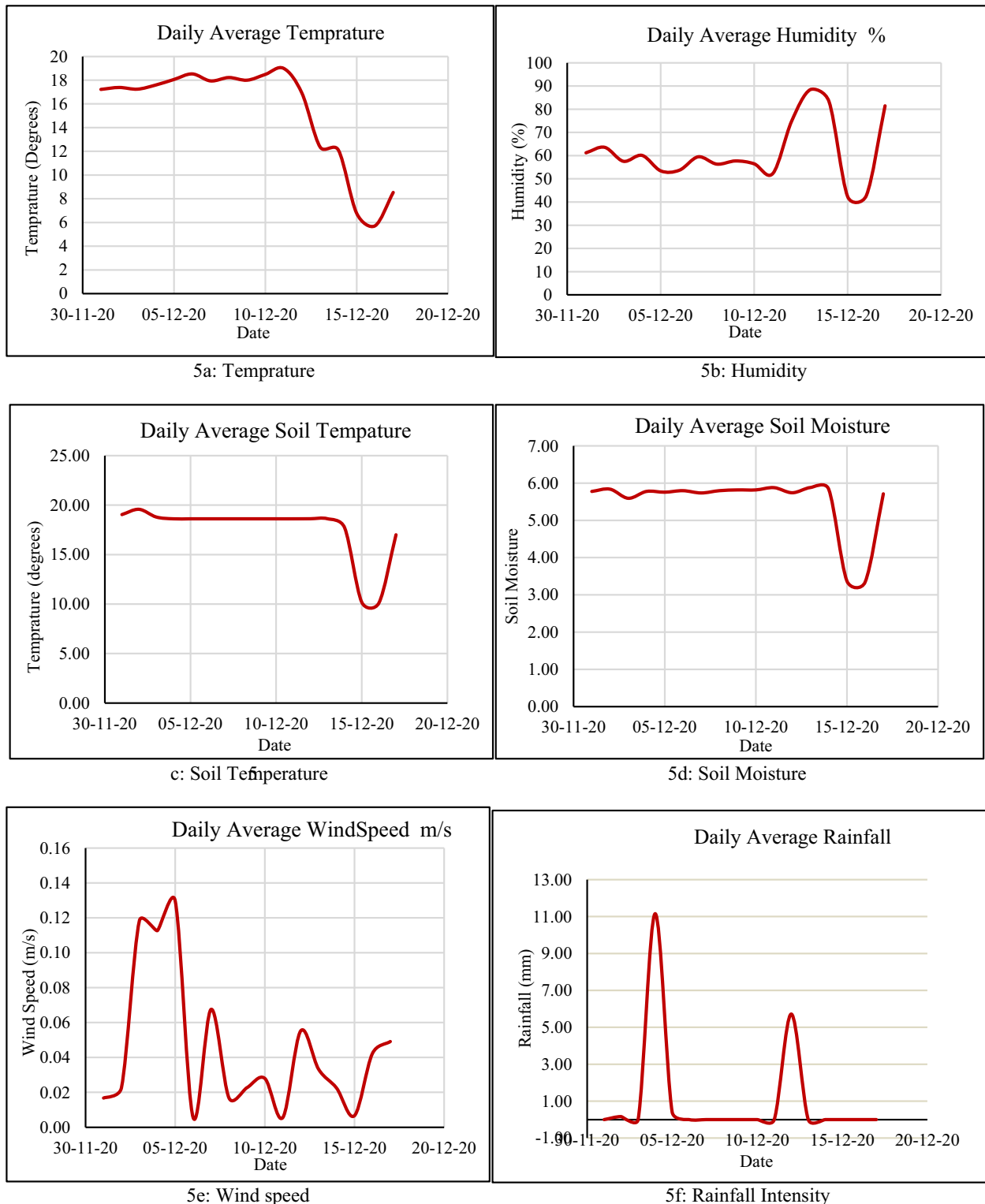


Fig. 5. (a) Temperature. (b) Humidity. (c) Soil Temperature. (d) Soil Moisture. (e) Wind speed. (f) Rainfall Intensity.

NGO's and residents. Visual method often provides high impacts during the warning period due to slope failure. We adopted three pronged categories for the display of visual messages (Table 1).

The test results for six different sensors have been provided between (01/12/2020 to 17/12/2020). The results derived for temperature and humidity corresponds with drop in temperature post 10/12/2020 and increased humidity. Drop in Soil tempera-

ture and moisture were observed since 15/12/2020 due to drop in surface temperature. Rainfall data were observed 24*7 with highest recorded amount 11.16 mm on 04/12/2020 (Fig. 5a to f). The sensors will further be checked and calibrated for threshold values and can be deployed in regions where stabilization of slopes are of high difficulty with respect to settlements and transportation corridors.

4. Conclusion

The methodology adopted in our research uses automated as well as knowledge driven method to evaluate the results of the data. During emergency times the datasets will be evaluated in-house through researchers and scientist before issuing the alert. The test results retrieved for the dates (01/12/2020 to 17/12/2020) for Temperature, humidity, soil moisture and temperature corresponds the daily average values. Rainfall data were observed 24*7 with highest recorded amount 11.16 mm on 04/12/2020. The sensors will be checked constantly and calibrated for threshold values and deployed in landside prone regions to form a part of real time networking and monitoring of landslides. In the field of disaster studies, landslide research and development of early warning system is of greater difficulty and paramount importance. The reason being same of set threshold values and sensors do not apply for every landslide. The sensors need to be calibrated for individual landslides. The research work presented can be used to monitor Debris type of landslides which is composed of Soil and Rock particles. The threshold values of various sensors can be adjusted manually post installation via remote network access. The sensors do not only address the issues of mass movements but also can be employed to address the rainfall intensity and soil movements along watershed catchments to predict potential landslide damming and flood scenarios.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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